

Predicting unplanned ward readmission from routine vital signs

A. Example, B. Sample, C. Fixture

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Abstract

We tested whether a simple logistic model using routine vital signs could flag adult inpatients at high risk of unplanned readmission within 30 days. The cohort was 1,842 discharges from one teaching hospital in 2023. The model used heart rate, respiratory rate, systolic blood pressure, temperature, oxygen saturation, and age. Discrimination on a held-out set of 461 discharges was modest, with an area under the receiver operating characteristic curve of 0.71. Calibration was acceptable in the middle of the risk range and poor above a predicted risk of 0.40. We conclude that vital signs alone are not enough for a deployment decision.

Introduction

Unplanned readmission is a common quality metric, but many scores need laboratory values or long notes that are missing at discharge. Electronic health record models have shown that rich clinical data can predict a range of outcomes, yet simpler inputs are easier to audit. Reporting guidance for prediction models now expects a clear target population, a missing-data plan, and a calibration assessment, not only a discrimination number. We asked a narrower question: do the six vital signs recorded in the last 24 hours before discharge separate patients who return within 30 days from those who do not?

Methods

We included adults discharged alive from general medicine between 1 January and 31 December 2023. The outcome was an unplanned inpatient stay at the same hospital starting within 30 days. Predictors were the last recorded heart rate, respiratory rate, systolic blood pressure, temperature, oxygen saturation, and age. We fit a logistic regression on 75 percent of discharges, chosen by calendar time, and evaluated it on the remaining 25 percent. Missing vital signs, present in 6 percent of rows, were filled with the training-set median. We did not tune a decision cutoff in the evaluation set. The protocol was fixed before that split was opened.

Results

Of 1,842 discharges, 214 (11.6 percent) were followed by an unplanned readmission. In the evaluation set of 461 discharges, the area under the receiver operating characteristic curve was 0.71 (95 percent interval 0.64 to 0.77). Sensitivity at a pre-specified score of 0.20 was 0.48 and specificity was 0.81. The calibration slope was 0.82. Patients with a predicted risk above 0.40 were too few for a stable estimate, and observed events in that band exceeded predictions. Age and an elevated respiratory rate had the largest coefficients. Oxygen saturation added little once respiratory rate was in the model.

Discussion

A vital-sign model with an area under the curve of 0.71 is not ready for ward use. The signal is real but small, the evaluation is internal, and high-risk predictions were miscalibrated. Guidance on artificial intelligence reporting would treat this as an early development study, not an implementation study. A useful next step is an external cohort and a decision analysis that states how many false alarms are acceptable per prevented readmission. We also did not measure whether clinicians would act on the score.

References

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